

Name: _____

Teacher: _____



HSC Mathematics

Extension 2

Assessment Task 2 - 2013

Time Allowed - 90 minutes +5 minutes reading

Instructions: Calculators may be used in any parts of the task. For 1 Mark Questions, the correct answer is sufficient to receive full marks. For Questions worth more than 1 Mark, necessary working MUST be shown to receive full marks.

Multiple Choice	/5
Question 6	/15
Question 7	/15
Question 8	/15
Total	/50

Answer Questions 1 to 5 on the separate multiple choice answer sheet.

Question 1

Consider the following statements about a polynomial $P(x)$:

- (i) If $P(x)$ is even, then $P'(x)$ is odd.
 (ii) If $P'(x)$ is even, then $P(x)$ is odd.

Which statement is always true

- A (i) only B (ii) only C Both (i) and (ii) D Neither (i) nor (ii)

Question 2

Let α, β, γ be the zeros of the polynomial $x^3 + 5x - 3$ The value of $\alpha^3 + \beta^3 + \gamma^3$ is

- A -125 B 0 C 9 D 34

Question 3

If $f(x) = \sqrt{1-x}$ then the common domain between $y = f(x)$ and $y = f(-x)$ is

- A $-1 \leq x \leq 1$ B $-1 < x < 1$ C $x \leq 1$ D $x < 1$

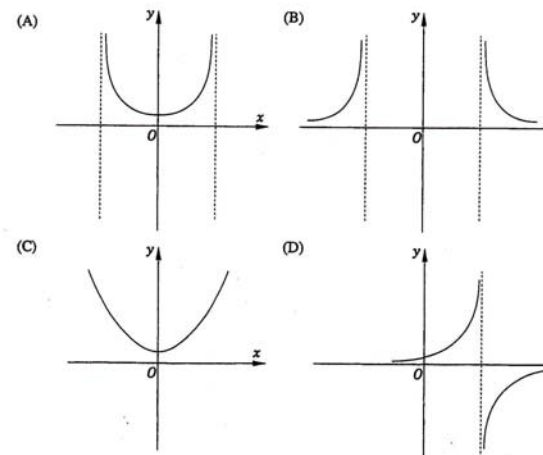
Question 4

Let $z = a + ib$ where $a \neq 0$ and $b \neq 0$

Which of the following statements is false?

- A $z - \bar{z} = 2bi$ B $|z|^2 = |z||\bar{z}|$ C $|z| + |\bar{z}| = |z + \bar{z}|$ D $\arg(z) + \arg(\bar{z}) = 0$

Question 5

Which of the following best represents the graph of $y = \frac{1}{\sqrt{4-x^2}}$?

Question 6 (15 Marks) Begin a new sheet of paper

Marks

a) Describe, give the equation and sketch the locus defined by

(i) $\text{Arg} [z - (1 + \sqrt{3}i)] = \frac{\pi}{3}$

2

(ii) $z^2 - \bar{z}^2 = 16i$

2

b) Let P, Q and R represent the complex numbers w_1, w_2 and w_3 respectively.

3

What geometric properties characterise triangle PQR if $w_2 - w_1 = i(w_3 - w_1)$.

Give reasons for your answer.

c) For the polynomial $P(x) = x^6 + x^3 + 1$

(i) Show that the roots of $P(x) = 0$ are amongst the roots of $x^9 - 1 = 0$

2

(ii) Hence show the roots of $P(x) = 0$ on the unit circle on an Argand diagram

3

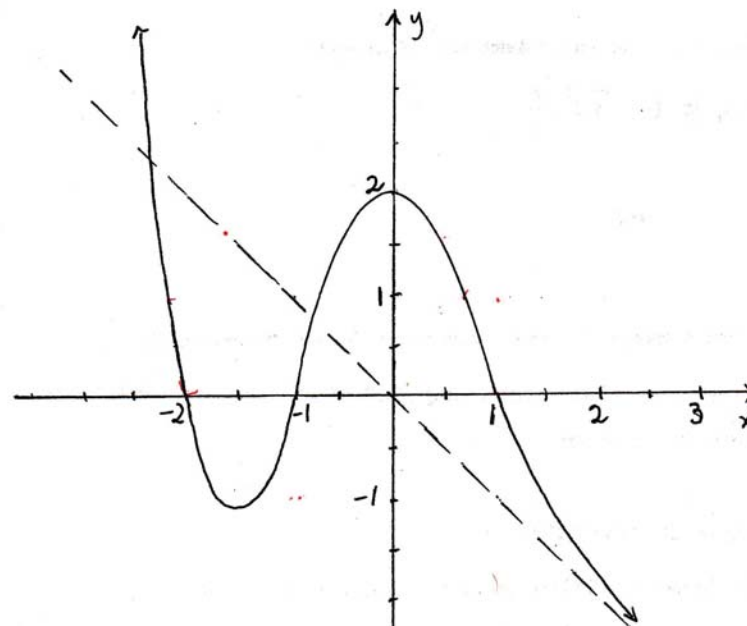
(iii) Hence show that

$$P(x) = (x^2 - 2x \cos \frac{2\pi}{9} + 1)(x^2 - 2x \cos \frac{4\pi}{9} + 1)(x^2 - 2x \cos \frac{8\pi}{9} + 1)$$

3

Question 7 (15 Marks) Begin a new sheet of paper

Marks



a) For the curve $y = f(x)$ given above, draw individual half page sketches showing intercepts, asymptotes, turning points and other important features of

(i) $y = [f(x)]^2$

2

(ii) $y^2 = f(x)$

2

(iii) $y = x f(x)$

2

(iv) $y = f(|x|)$

2

(v) $y = \ln f(x)$

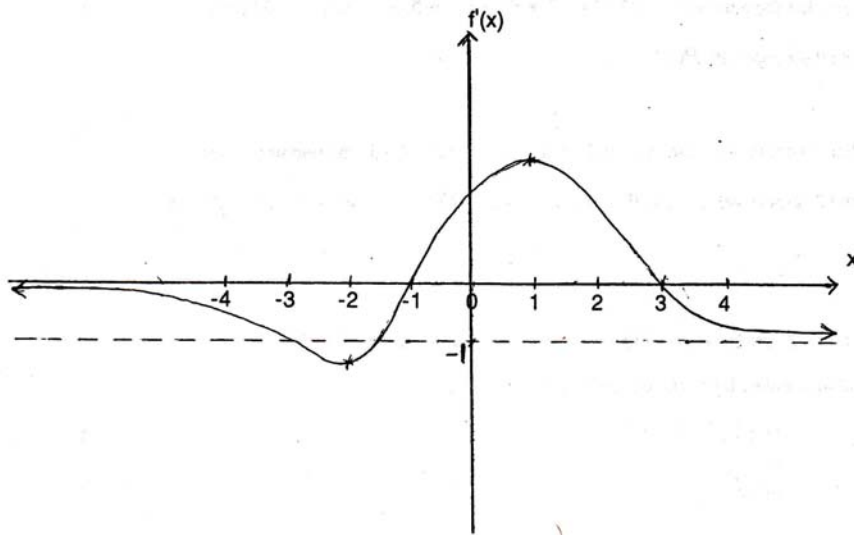
2

b) Part b) is on the next page.

Question 7 continued

Marks

- b) The function $y = f(x)$ has a derivative $y = f'(x)$ whose graph is given:



Sketch $y = f(x)$ given that $f(0) = 0$ and $f(-3) > 0$.

3

Describe the behaviour of $y = f(x)$ as $x \rightarrow \infty$ and as $x \rightarrow -\infty$.

2

Question 8 (15 Marks) Begin a new sheet of paper

Marks

- a) Given that the polynomial $P(x) = x^4 + x^3 - 3x^2 - 5x - 2$ has a 3 fold root, find all the roots of $P(x)$ 3
- b) When a certain polynomial is divided by $x+1$ and $x-3$, the remainders are 6 and 2 respectively. Find the remainder when $P(x)$ is divided by $x^2 - 2x - 3$. 3
- c) If α, β, γ are the roots of the cubic equation $2x^3 - 3x + 10 = 0$. Find in simplest form, the equation with roots:
- i) $\alpha + 1, \beta + 1, \gamma + 1$ 2
- ii) $\alpha^3, \beta^3, \gamma^3$ 3
- d) Consider the equation $z^3 + mz^2 + nz + 6 = 0$, where m and n are real. It is known that $1 - i$ is a root of the equation.
- i) Find the other two roots of the equation. 2
- ii) Find the values of m and n . 2

END OF TEST



Name: _____

Teacher: _____

Multiple-choice answer sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely, using a black pen.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct
 ↓

Start here → 1. A ☐ B ☐ C ☐ D ☐

2. A ☐ B ☐ C ☐ D ☐

3. A ☐ B ☐ C ☐ D ☐

4. A ☐ B ☐ C ☐ D ☐

5. A ☐ B ☐ C ☐ D ☐

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, x > 0$

Solutions 2013 Set 2 Task 2

Q1 i) true ii) $P(x)$ has a constant added $\therefore$ false. (A)

Q2 Consider $x^3 + 5x - 3 = 0$

$$x^3 = 3 - 5x$$

$$\alpha^3 = 3 - 5\alpha$$

$$\beta^3 = 3 - 5\beta$$

$$\gamma^3 = 3 - 5\gamma$$

$$\alpha^3 + \beta^3 + \gamma^3 = 9 - 5(\alpha + \beta + \gamma) \quad (C)$$

$$= 9 - 0$$

Q3 Domain of $f(x)$: $x \leq 1$
 $f(-x) = \sqrt{1+x}$ D: $x \geq -1$
 common domain $-1 \leq x \leq 1$ (A)

Q4 $|z| + |\bar{z}| = \sqrt{a^2+b^2} + \sqrt{a^2+b^2}$; $|z+\bar{z}| = 2a$ (C)

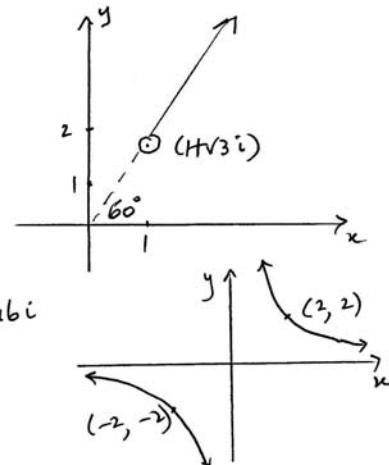
Q5 Domain $4-x^2 > 0 \Rightarrow -2 < x < 2$ (A)

Question 6

a) i) $\text{Arg}[z - (1+\sqrt{3}i)] = \frac{\pi}{3}$

Locus is a ray beginning at $(1+\sqrt{3}i)$ + heading away from 0.

Equations: $y = \sqrt{3}x$, $x > 1$



ii) $(x+iy)^2 - (x-iy)^2 = 16i$
 $(x^2 - y^2 + 2xyi) - (x^2 - y^2 - 2xyi) = 16i$

$$4xyi = 16i$$

$$xy = 4$$

Locus is a hyperbola in quadrants (1) + (3)

Equation $xy = 4$

b) $\triangle PQR$ is right angled at $P(w_1)$

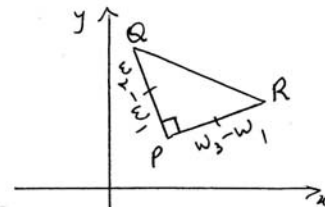
and is isosceles $PQ = PR$.

Multiplication of $(w_3 - w_1)$ by i rotates this vector 90° anticlockwise + does not change its length since

$$|w_2 - w_1| = |i(w_3 - w_1)|$$

$$= |i| |w_3 - w_1|$$

$$= |w_3 - w_1|$$



c) i) $x^9 - 1 = (x^3)^3 - 1$ using $x^3 - y^3 = (x-y)(x^2 + xy + y^2)$

Factorising $= (x^3 - 1)(x^6 + x^3 + 1)$

$$\therefore x^9 - 1 = 0 = (x^3 - 1)(x^6 + x^3 + 1)$$

$$x^3 - 1 = 0$$

$$x^6 + x^3 + 1 = 0$$

Cube roots of unity remaining roots of the equation

$\therefore$ Roots of $x^6 + x^3 + 1 = 0$ are all the roots of

$x^9 - 1 = 0$ which are not cube roots of 1.

ii) Solving $x^9 = 1$

$$(\cos\theta + i\sin\theta)^9 = 1 \cos 0$$

$$\cos 9\theta = 1 \cos 0$$

ii)

$$9\theta = 0 + 2n\pi$$

$$n=0$$

$$\theta = 0$$

$$x_0 = \text{cis } 0 = 1$$

$$n=1$$

$$\theta = \frac{2\pi}{9}$$

$$x_1 = \text{cis } \frac{2\pi}{9}$$

$$n=2$$

$$\theta = \frac{4\pi}{9}$$

$$x_2 = \text{cis } \frac{4\pi}{9}$$

$$n=3$$

$$\theta = \frac{6\pi}{9}$$

$$x_3 = \text{cis } \frac{6\pi}{9} = \text{cis } \frac{2\pi}{3}$$

$$n=4$$

$$\theta = \frac{8\pi}{9}$$

$$x_4 = \text{cis } \frac{8\pi}{9}$$

$$n=5$$

$$\theta = \frac{10\pi}{9}$$

$$x_5 = \text{cis } \frac{10\pi}{9} = \text{cis } -\frac{8\pi}{9}$$

$$n=6$$

$$\theta = \frac{12\pi}{9}$$

$$x_6 = \text{cis } \frac{12\pi}{9} = \text{cis } \frac{4\pi}{3}$$

$$n=7$$

$$\theta = \frac{14\pi}{9}$$

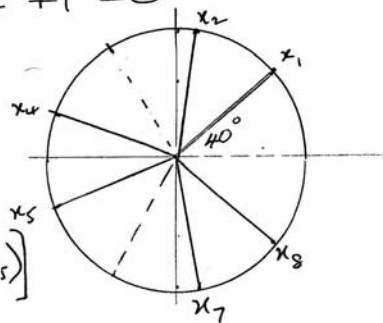
$$x_7 = \text{cis } \frac{14\pi}{9} = \text{cis } -\frac{4\pi}{9}$$

$$n=8$$

$$\theta = \frac{16\pi}{9}$$

$$x_8 = \text{cis } \frac{16\pi}{9} = \text{cis } -\frac{2\pi}{9}$$

Now x_0, x_3, x_6 are cube roots of unity
+ are not roots of $x^6 + x^3 + 1 = 0$



$$\text{iii) } P(x) = (x-x_1)(x-x_2)(x-x_4)(x-x_5)(x-x_7)(x-x_8)$$

$$= [(x-x_1)(x-x_8)][(x-x_2)(x-x_7)][(x-x_4)(x-x_5)]$$

$$= \left[\left(x - \text{cis } \frac{2\pi}{9} \right) \left(x - \text{cis } -\frac{2\pi}{9} \right) \right] \left[\left(x - \text{cis } \frac{4\pi}{9} \right) \left(x - \text{cis } -\frac{4\pi}{9} \right) \right] \left[\left(x - \text{cis } \frac{8\pi}{9} \right) \left(x - \text{cis } -\frac{8\pi}{9} \right) \right]$$

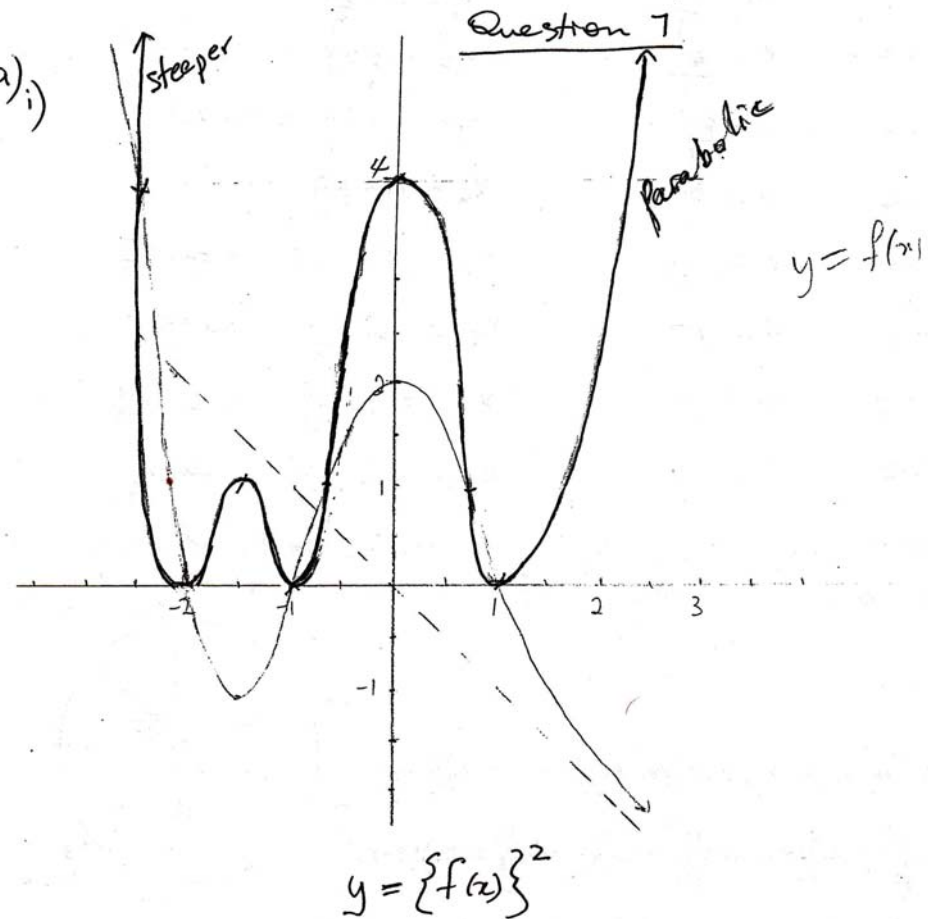
$$= \left(x^2 - 2x \cos \frac{2\pi}{9} + 1 \right) \left(x^2 - 2x \cos \frac{4\pi}{9} + 1 \right) \left(x^2 - 2x \cos \frac{8\pi}{9} + 1 \right)$$

$$\text{as } (x-z)(x-\bar{z}) = x^2 - (z+\bar{z})x + z\bar{z}$$

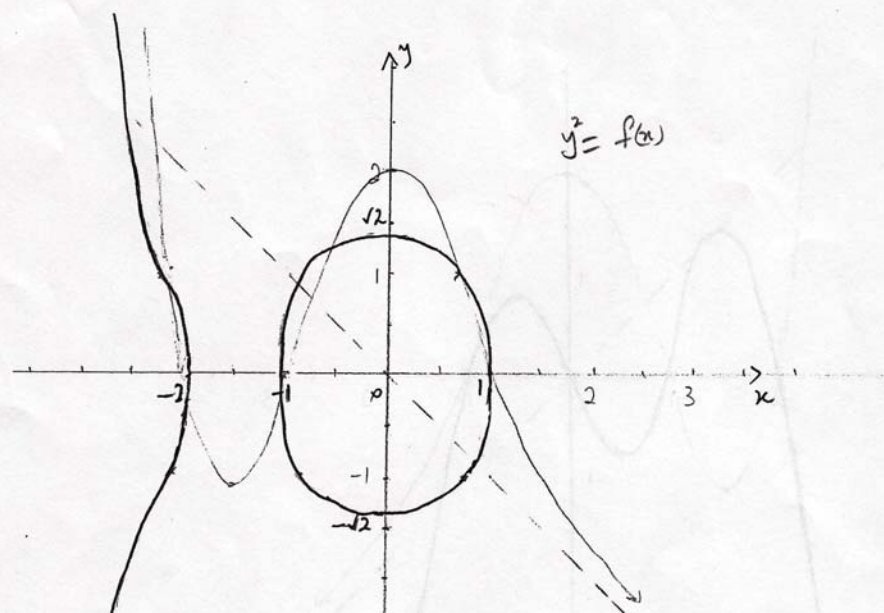
$$z+\bar{z} = 2a = 2 \cos \frac{n\pi}{9}$$

$$z\bar{z} = |z|^2 = 1$$

a) i)



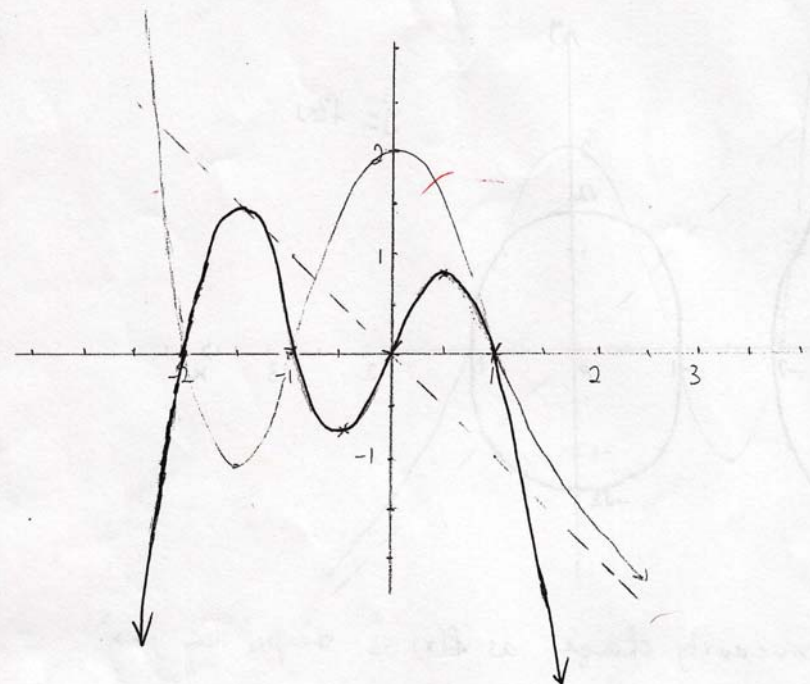
ii)



$$y^2 = f(x)$$

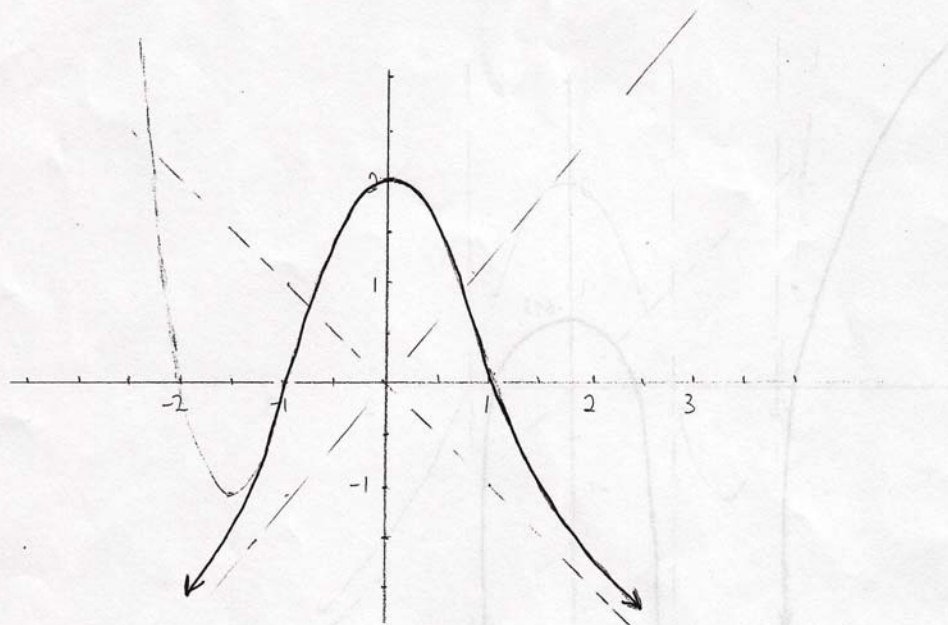
concavity change as $f(x)$ is steeper than $y = x^2$

iii)



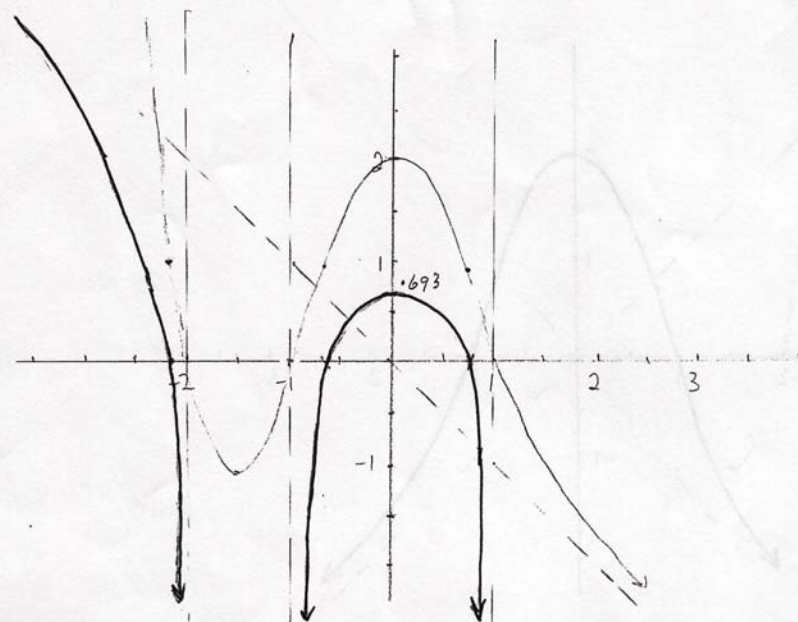
$$y = x f(x)$$

iv)



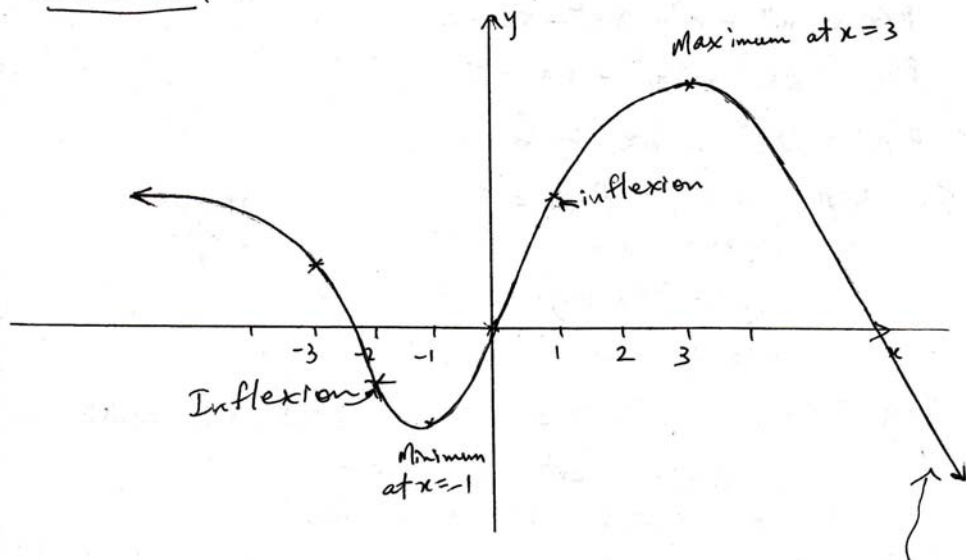
$$y = f(x)$$

v)



$$y = \ln\{f(x)\}$$

Question 7 b)



As $x \rightarrow \infty$ curve approaches line of gradient -1

As $x \rightarrow -\infty$ curve approaches $y = c$ where c is a positive constant as $f(-3) > 0$ + there are no more stationary points or inflexions.

Question 8

$$P(x) = x^4 + x^3 - 3x^2 - 5x - 2$$

$$P'(x) = 4x^3 + 3x^2 - 6x - 5$$

$$P''(x) = 12x^2 + 6x - 6$$

For triple root, $P''(x) = 0$

$$6(2x^2 + x - 1) = 0$$

$$(2x-1)(x+1) = 0$$

$$x = \frac{1}{2} \text{ or } -1$$

$$\begin{array}{r} 2x-1 \\ x+1 \\ \hline \end{array}$$

Try $P(-1)$ $P(-1) = 1 - 1 - 3 + 5 - 2 = 0$

$\therefore$ Triple root is $x = -1$

or $(x+1)^3$ is a factor

By inspection remaining factor is $(x-2)$

$$\therefore P(x) = (x+1)^3(x-2)$$

roots are $x = -1$ (triple) + $x = 2$

b)

$$P(-1) = 6$$

$$P(3) = 2$$

$$\text{Let } P(x) = (x^2 - 2x - 3)Q(x) + R(x)$$

$$P(x) = (x+1)(x-3)Q(x) + ax+b$$

$R(x) = ax+b$ as it must be of degree less than 2.

$$P(-1) = 6$$

$$6 = -a + b \quad \text{--- (1)}$$

$$P(3) = 2$$

$$2 = 3a + b \quad \text{--- (2)}$$

$$\text{(2) - (1)}$$

$$-4 = 4a \Rightarrow a = -1$$

$$b = 6$$

$\therefore$ Remainder is $-x + 6$.

8 c) $P(x) = 2x^3 - 3x + 10 = 0$

i) Put $y = x+1 \Rightarrow x = y-1$

Equation is $2(y-1)^3 - 3(y-1) + 10 = 0$
 $2\{y^3 - 3y^2 + 3y - 1\} - 3y + 3 + 10 = 0$
 $2y^3 - 6y^2 + 3y + 11 = 0$

ii) Put $y = x^3 \Rightarrow x = \sqrt[3]{y}$

$2y - 3y^{1/3} + 10 = 0$

$2y + 10 = 3y^{1/3}$

$(2y+10)^3 = 27y$

$8y^3 + 3 \times 4y^2 \times 10 + 3 \times 2y \times 100 + 1000 = 27y$

$8y^3 + 120y^2 + 600y + 1000 = 27y$

$8y^3 + 120y^2 + 573y + 1000 = 0$

d) $z^3 + mz^2 + nz + 6 = 0$

$1-i$ is a root. Coefficients are real $\therefore 1+i$ is root

$\therefore$ Product of roots $(1-i)(1+i) \cdot \alpha = -6$

$2\alpha = -6$

$\alpha = -3$

roots are $(1-i)$, $(1+i)$, -3 .

ii) Sum of roots $1-i + 1+i - 3 = -m$

$-1 = -m$

$m = 1$

$\therefore z^3 + z^2 + nz + 6 = 0$

Sub $z = -3$

$-27 + 9 - 3n + 6 = 0$

$-12 = 3n$

$-4 = n$

$\therefore m = 1 \text{ and } n = -4$